Submission Worksheet

Submission Data

Course: IT490-451-M2025
Assignment: IT490 MQ Test Individual
Student: Omar B. (ob75)
Status: Submitted | **Worksheet Progress:** 100%
Potential Grade: 10.00/10.00 (100.00%)
Received Grade: 0.00/10.00 (0.00%)
Started: 6/12/2025 9:35:27 PM
Updated: 6/12/2025 11:19:06 PM
Grading Link: https://learn.ethereallab.app/assignment/v3/IT490-451-M2025/it490-mq-test-individual/grading/ob75
View Link: https://learn.ethereallab.app/assignment/v3/IT490-451-M2025/it490-mq-test-individual/view/ob75

Instructions

- Walkthrough: https://youtu.be/tgT0ZAxccbQ

1. Read all instructions and requirements first
2. Use any VM creation tool that gives you root access and persistent storage
   - VirtualBox, Multipass, cloud (Amazon, Google, Azure, etc) (Docker won't be an option here)
   - Create a hostname relevant to the assignment (i.e., test-individual)
   - Create a user of your ucid with a password, ensure relevant permissions
   - Hardward: 1GB Memory, 10GB Hard Drive
   - Install a server version of linux (i.e., Ubuntu Server 24.04)
   - Hint: You may want to get a base install working and use that as a cloning point for quicker destroy/create cycles
3. Use the example code from the master branch of https://github.com/MattToegel/IT490
4. Connect to the VM with two separate ssh connections
   - Run the RabbitMQServerSample.php file successfully in one instance
   - Run the RabbitMQClientSample.php file successfully in another instance
   - Proper data should be sent/received
5. Create a `setup.sh` script that automates the installation/setup logic
6. Fill in the below requirements
7. Submit and Export once done
8. Upload the PDF to your personal GitHub repo for the class
9. Upload the PDF to Canvas

Section #1: (7 pts.) Example Solution

Progress: 100%

Task #1 (3.50 pts.) - Working Example

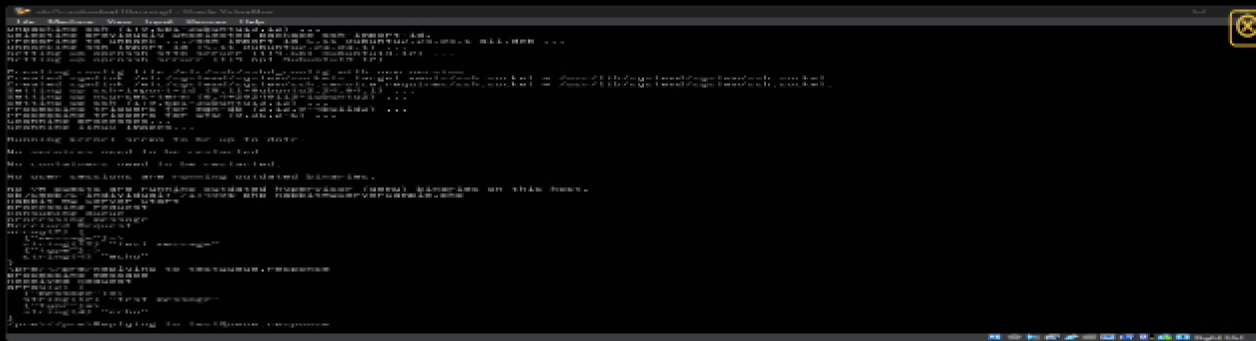
Progress: 100%

Part 1:

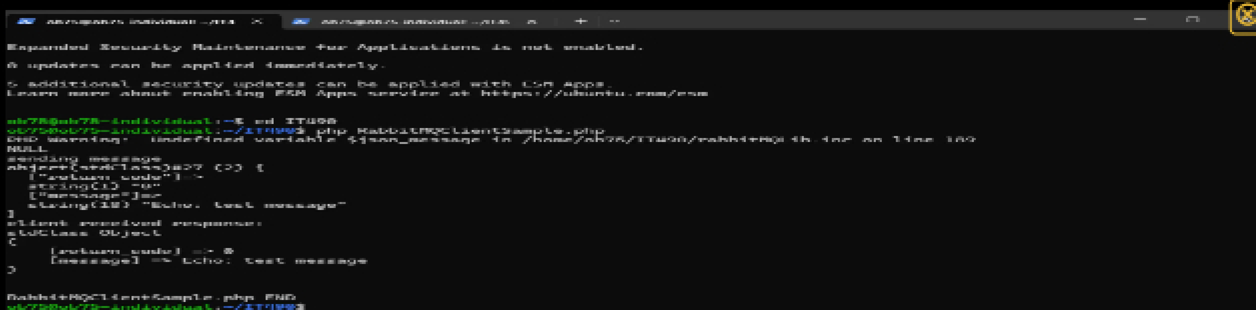
Progress: 100%

Details:

- Demonstrate a successful send/receive of the example message
- Hostname should be test-individual or similar
- Username should be your ucid



Screenshot showing the server consuming and replying to the client.



Screenshot showing client sending and receiving messages.

Screenshot showing 2 ssh connections to the RabbitMQServer.

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Part 2:

Progress: 100%

Details:

- Detail the initial setup experience and note things you had to address in order for the example to work

Your Response:

During the initial setup process for this to work, I had to first do sudo apt update and sudo apt upgrade to make sure that the server has the references updated and also install any updates that maybe available to the machine so that everything can be up to date. Also had to install certain other commands such as php, composer, and rabbitmq-server. These would allow us to use the commands to use our php files. Then the repository with the example code is cloned onto the machine to access the files directly and run the rabbitmq-server file. The type parameter was invalid in the composer.json file so I had to lower case the words in the type parameter and it was able to open the file. Then the RabbitMQServerSample.php file was started and waited for responses from the client. Once the client connects the server consumes and replies to the client.

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Task #2 (3.50 pts.) - Setup Script

Progress: 100%

Part 1:

Progress: 100%

Details:

- Show a snippet of the `setup.sh` script you created to automate the installation and configuration steps that lead up to a working example.

Screenshot showing the setup.sh file process to automate the startup/installation process.

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Part 2:

Progress: 100%

Details:

- Include the direct link to the file from your personal class repository

URL #1

https://github.com/OmarBarrera1/ob75-IT490-Individual/blob/main/setup.sh

URL

https://github.com/OmarBarrera1,

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Part 3:

Progress: 100%

Details:

- Briefly explain each step of the process in the script

Your Response:

First what was done was the shebang line that allows all the lines in the script to be executable. Then set -e will exit the shell if there is an error and then we want to update and upgrade our references on our machine so we do sudo apt update as well as sudo apt upgrade -y. Then we install our composer and rabbitmq-server to be able to run our rabbitmq server scripts from the examples. We clone the repository to fetch the exmaples by calling the repository after git clone and then we cd into the main branch after we got the repository. Next step, we use jq to work with JSON commands and interact with them and we want to lowercase any of the words that are found in "type" since that will give us a JSON error when we run the files and its uppercase. We then update our composer and run our php RabbitMQServer.php file and that loads the server up and waits for a message.

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Section #2: (3 pts.) Reflection

Progress: 100%

Task #1 (1 pt.) - What was the easiest part of this assignment

Progress: 100%

Details:

- At least a few solid sentences

Your Response:

The easiest part of the assignment was setting up the VM with the Ubuntu server to be loaded onto the machine. Another simple part was installing all the packages necessary to use the commands that were needed to open the files for the assignment such as sudo apt update, sudo apt install composer, etc. Finally cloning the repository to also use the files was also simple to be able to access the files.

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Task #2 (1 pt.) - What was the hardest part of this assignment

assignment

Progress: 100%

Details:

- At least a few solid sentences

Your Response:

The hardest part of the assingment was setting up a connection so that I can log into to VM from the powershell so that I can open the client sample of rabbitMQ to connect to the server. I needed to setup a port number 2222 on host port and set guest port to 22 and set it up over TCP. This allowed my machine to have a connection and I also needed to install ssh by doing sudo apt install ssh.

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Task #3 (1 pt.) - What did you learn during this assignment

Progress: 100%

Details:

- At least a few solid sentences

Your Response:

During this assignment, I learned the sudo commands that were necessary to download the packages and references we needed. I also learned on how to have a virtual machine running and have other terminals connect to that virtual machine remotely via another terminal window and taking control of the machine and setting up a client.

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